

Homework 1 Solutions

Problem 1

In each of the following, write a clearly worded sentence interpreting the numerical value of the probability as a long run relative frequency in context. (Just take the numerical values as given for now. We'll see how to compute probabilities like these later.)

- a. The probability of rolling doubles when you roll two fair six-sided dice is $1/6$.
- b. The probability of rolling doubles on three consecutive rolls of two fair six-sided dice is 0.00463.
- c. The probability that the sum of 100 rolls of a fair six-sided die is less than 370 is 12%.
- d. Roll a fair six-sided die until you roll a 6 three times and then stop. The probability that you roll the die at least 10 times is 0.822.

Solution

- a. In about 16.7% sets of two rolls both rolls in the set are 6.
- b. In about 0.463% of sets of three rolls all three rolls are 6. (If the sets consisted of more rolls, say 10 rolls, the probability of at least 3 consecutive 6s somewhere in the set would be greater.)
- c. In about 12% of sets of 100 rolls the sum of the 100 rolls is less than 370.
- d. In about 82.2% of sets of 9 rolls there are at most 2 6s.

Problem 2

1. Suppose that at some point in 2023 your subjective probabilities for who would win the 2024 U.S. Presidential Election satisfied the following. (Imagine a time before Trump had been declared the Republican nominee, and while Biden was the presumptive Democratic nominee there was still speculation that he should step aside.)

- The Democratic nominee and the Republican nominee are equally likely to be the winner of the Presidential Election
- Joe Biden is 5 times more likely to win than Kamala Harris, and no other Democratic candidate has a chance of winning
- Donald Trump is twice as likely to win as any other Republican candidate.

Create a table of your subjective probabilities.

Solution

It helps to designate one outcome as the “baseline”. It doesn’t matter which one; we’ll choose Kamala Harris.

- Suppose Harris accounts for 1 “unit”.
- Biden accounts for 5 units
- Other Democrat candidates account for 0 units
- Democrats in total account for 6 units
- Republicans in total account for 6 units
- Trump accounts for 4 units
- Other Republican accounts for 2 units

Candidate	“Units”	Probability	As Percent
Biden	5	$5/12 = 0.417$	41.7%
Harris	1	$1/12 = 0.083$	8.3%
Other Democrat	0	0	0%
Trump	4	$4/12 = 0.333$	33.3%
Other Republican	2	$2/12 = 0.167$	16.7%
Total	12	1	100%

Problem 3

Suppose that you have applied to two graduate schools, A and B. Your subjective probability of being accepted is 0.6 for school A and 0.7 for school B. Hint: the fact that these are subjective probabilities does not change the way you solve the problem. Remember, the mathematics of probability is the same regardless of whether the probabilities represent long run relative frequencies or subjective degrees of relative likelihood. Just take the 0.6 and 0.7 values as given and use them to solve the following parts.

- Interpret the subjective probabilities: How many times more likely than not are you to be accepted at school A? How many times more likely are you to be accepted at school B than at school A?

- b. What is the largest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
- c. What is the smallest possible probability of being accepted by both schools? Under what scenario (however unrealistic) would this be true? Explain.
- d. Explain why your subjective probability of being accepted by both schools is not necessarily 0.42.
- e. **For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55.** If you are accepted at school A, what is your probability of also being accepted at school B? For this and the remaining parts in addition to computing the probability represent it with proper notation.
- f. If you are accepted at school A, what is your probability of not being accepted at school B?
- g. If you are not accepted at school A, what is your probability of being accepted at school B?
- h. If you are accepted at school B, what is your probability of also being accepted at school A?
- i. If you are not accepted at school B, what is your probability of being accepted at school A?

Solution

- a. The probability of being accepted at both schools can be no more than the probability of being accepted at either school, 0.6 and 0.7. So we try 0.6 and see if we get valid probabilities.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.6	0.1	0.7
Not accepted by B	0	0.3	0.3
Total	0.6	0.4	1.00

All the probabilities above are valid, so the largest the probability of being accepted at both schools can be is 0.6. This occurs if getting into school A guarantees that you will also get into school B.

- b. Trying a subjective probability of being accepted at both schools leads to a negative probability of not being accepted at either school. So try a probability of 0 for the probability of not being accepted at either school.

	Accepted by A	Not accepted by A	Total
Accepted by B	0.3	0.4	0.7
Not accepted by B	0.3	0	0.3

	Accepted by A	Not accepted by A	Total
Total	0.6	0.4	1.00

So the smallest the probability of being accepted at both schools can be is 0.3. This only occurs if you're guaranteed of getting into at least one of the two schools.

- c. Your chances of getting into one school might change depending if you've gotten into the other. For example, your chances of getting to Cal Poly are probably higher if you know you've already gotten into Stanford.
- d. For the remaining parts, suppose your subjective probability of being accepted at both schools is 0.55. If you are accepted at school A, what is your probability of also being accepted at school B?

	Accepted by A	Not accepted by A	Total
Accepted by B	0.55	0.15	0.70
Not accepted by B	0.05	0.25	0.30
Total	0.60	0.40	1.00

If you are accepted at school A, then your probability of also being accepted at school B is $P(B|A) = 0.55/0.60 = 0.917$. (91.7% of students like you who get into school A also get into school B.)

- e. If you are accepted at school A, what is your probability of not being accepted at school B?

If you are accepted at school A, then your probability of not being accepted at school B is $P(B^c|A) = 0.05/0.60 = 0.083$. (8.3% of students like you who get into school A do not get into school B.)

- f. If you are not accepted at school A, what is your probability of being accepted at school B?

If you are not accepted at school A, then your probability of not being accepted at school B is $P(B|A^c) = 0.15/0.40 = 0.375$. (37.5% of students like you who do not get into school A do get into school B.)

- g. If you are accepted at school B, what is your probability of also being accepted at school A?

If you are accepted at school B, then your probability of being accepted at school A is $P(A|B) = 0.55/0.7 = 0.786$. (78.6% of students like you who get into school B also get into school A.)

- h. If you are not accepted at school B, what is your probability of being accepted at school A?

If you are not accepted at school B, then your probability of being accepted at school A is $P(A|B^c) = 0.05/30 = 0.167$. (16.7% of students like you who do not get into school B do get into school A.)

Problem 4

For each of the following, create your own “which of (a) or (b) is larger?” example. You can use the ones from the handout as examples, but you should choose your own contexts. Identify the correct answer and write a sentence or two explaining the answer to a student who is confused. You can use illustrations in your explanations, but don’t do any calculations.

- Reverse the direction of conditioning (like the 6 foot tall and NBA example)
- Compare conditional and unconditional probabilities (like man versus man who is greater than 6 feet tall)
- Two “particulars” that people might think are not equally likely but are (like HHHHH versus HTHTT).
- Comparing “the particular” and “the general” (HTHTT versus 2 H in 5 flips).
- Comparing the probability of happening once versus at least once (like you winning the lottery versus someone winning the lottery)

Solution

Results vary. See [Section 1.6 of my probability textbook](#) for discussion.

Problem 5

The “matching problem” involves n distinct objects labeled $1, \dots, n$ which are placed in n distinct boxes labeled $1, \dots, n$, with exactly one object placed in each box. Suppose the objects are placed in the boxes uniformly at random, so that any possible arrangement is equally likely. We’re interested in the number of matches (the number of objects for which their label matches the label of the box in which they are placed.) Let C be the event that at least one object is placed in the correct spot (at least one match).

- Describe in full detail how, in principle, you would use physical objects (like cards) to simulate a single repetition of the placement of objects in boxes, and the number of matches.
- Describe in full detail how, in principle, you could conduct a simulation and use the results to approximate $P(C)$.

- c. [This applet conducts a simulation of the matching problem](#) (in the context of babies in a hospital being mixed up and returned to parents uniformly at random.) In the applet, “Number of babies” represents n and “Number of trials” represents the number of simulated repetitions. Starting with $n = 4$, try different values for the number of repetitions and hit the “Randomize” button to run the simulation. It helps to start with just a few repetitions so that you can see how the simulation is working, before jumping to thousands of repetitions. Use the simulation results to approximate $P(C)$ when $n = 4$.
- d. How do you expect $P(C)$ to depend on n ? As n increases, do you expect $P(C)$ to increase, decrease, or stay about the same? Just think about it before proceeding; what does your intuition say?
- e. Now try a few different values of n (say $n = 5$, $n = 10$, and $n = 20$, but you’re free to choose other values). For each value of n run the simulation and use the results to approximate $P(C)$. Record your results.
- f. What do the simulation results suggest about the relationship between $P(C)$ and n ? (Remember that there is simulation margin of error, based on the number of repetitions. Two numerical approximations that are within the margin of error of each other are essentially the same approximation.)

Solution

- a. Use a box with 4 tickets, labeled 1, 2, 3, 4. Shuffle the tickets and draw all 4 *without* replacement and record the tickets drawn *in order*. Let X be the number of tickets that match their spot in order. (For example, if the tickets are drawn in the order 2431 then the realized value of X is 1 since only ticket 3 matches its spot in the order.) Since $C = \{X \geq 1\}$, event C occurs if $X \geq 1$ and does not occur if $X = 0$. We could record the realization of event C as “True” or “False”.
- b. Repeat the previous part many times. Count the number of repetitions on which C occurs and divide by the total number of repetitions to approximate $P(C)$.
- c. See the applet.
- d. You might think the probability might decrease with n , since the probability that any particular object goes in its correct spot decreases with n . On the other hand, as n increases, the more chances there are to get at least one object in the correct spot. These considerations move in opposite directions; what happens as n increases?
- e. When $n = 4$, $P(X = 0)$ is approximately 0.375; when $n = 5$, $P(X = 0)$ is approximately 0.367; but for $n \geq 6$, $P(X = 0)$ is approximately 0.368 for any value of n .
- f. The probability of no matches essentially does not depend on n ! We’ll investigate a little more in class.

Problem 6

Katniss throws a dart at a circular dartboard with radius 1 foot. Suppose that Katniss's dart lands at a uniformly random location on the dartboard (and she never misses the dartboard).

- Compute the probability that Katniss's dart lands within 1 inch of the center of the dartboard.
- Compute the probability that Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

Solution

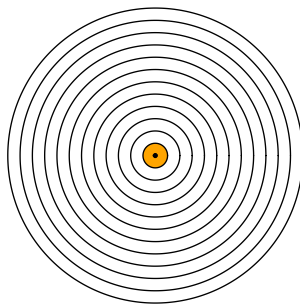
See Figure 24.1 for pictures.

- Figure 24.1a: A , Katniss's dart lands within 1 inch of the center of the dartboard.
- Figure 24.1b: B , Katniss's dart lands more than 1 inch but less than 2 inches away from the center of the dartboard.

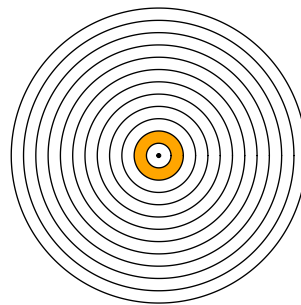
Since the dart lands uniformly at random anywhere on the dartboard, probabilities are computed as the ratio between the area corresponding to the event of interest divided by the area of the total dartboard ($12^2\pi$ square-inches.)

Find the area of the shaded pieces by subtracting areas of circles.

- $P(A) = \frac{1^2\pi}{12^2\pi} = 1/144 = 0.00694$
- $P(B) = \frac{2^2\pi - 1^2\pi}{12^2\pi} = 3/144 = 0.021$



(a) Within 1 inch of center



(b) More than 1 but less than 2 inches from center

Figure 24.1: Events in dartboard problem

Problem 7

1. Imagine a light that flashes every few seconds³. The light randomly flashes green with probability 0.75 and red with probability 0.25, independently from flash to flash.
 - a. Write down a sequence of G's (for green) and R's (for red) to predict the colors for the next 40 flashes of this light. Before you read on, please take a minute to think about how you would generate such a sequence yourself.
 - b. Most people produce a sequence that has 30 G's and 10 R's, or close to those proportions, because they are trying to generate a sequence for which each outcome has a 75% chance for G and a 25% chance for R. That is, they use a strategy in which they predict G with probability 0.75, and R with probability 0.25. How well does this strategy do? Compute the probability of correctly predicting any single item in the sequence using this strategy.
 - c. Describe a better strategy. (Hint: can you find a strategy for which the probability of correctly predicting any single flash is 0.75?)

Solution

- a. Most people write a sequence with about 30 G and 10 R.
- b. Use the law of total probability. There are two cases for your guesses: G or R.
 - If you guess G then you are correct if it really flashes G, and it flashes G with probability 0.75.
 - If you guess R then you are correct if it really flashes R, and it flashes R with probability 0.25.

If you use a strategy where you guess G with probability 0.75, your probability of being correct is

$$\begin{aligned} P(\text{correct}) &= P(\text{correct}|\text{guess G})P(\text{guess G}) + P(\text{correct}|\text{guess R})P(\text{guess R}) \\ &= (0.75)(0.75) + (0.25)(0.25) = 0.625. \end{aligned}$$

- c. If you get predict G every time, you are correct any time it is G, so the probability that you are correct is 0.75.

³Thanks to Allan Rossman for this example.